Selecting for Stability

Ensemble Proximity as a Criterion for Lower
Out-of-Sample Prediction Variance

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Abstract

Neural networks trained with different random seeds produce models with similar accuracy but systematically different predictions on individual examples. This prediction variance is a nuisance in deployment settings that value consistency. We study *ensemble proximity* as a model selection criterion: among models with comparable accuracy, choose the one whose predictions most closely resemble the ensemble mean. Using a factorial simulation over 22 data-generating process configurations varying class separation, label noise, number of classes, sample size, and feature redundancy, we find that ensemble proximity selection reduces out-of-sample prediction variance by a mean of 7.4% (median 7.6%, 95% CI $[-0.7\%, +14.9\%]$), with reductions of 15–20% on hard problems, at essentially zero accuracy cost (mean gap: +0.004 percentage points). The benefit is conditional: it scales with baseline prediction disagreement among models and is negligible when models already agree. Validation-set proximity to the ensemble is a reliable predictor of test-set proximity (mean Spearman $\rho = 0.50$, significant in 41% of configurations), confirming the criterion transfers out of sample. Ensemble proximity outperforms alternative selection criteria including random selection, calibration-based selection, and entropy-based selection.

1 Introduction

Training a neural network involves several sources of randomness: weight initialization, minibatch ordering, data augmentation, and dropout masks. Two networks with identical architectures, hyperparameters, and training data generally converge to different solutions. Despite this, their aggregate performance metrics—validation accuracy, area under the ROC curve—are often similar, within a few tenths of a percentage point for well-tuned architectures.

This creates a practical selection problem. Given K trained models with roughly equivalent accuracy, which should be deployed? The default is to pick the one with the highest validation accuracy. But this optimizes for the wrong thing if the practitioner cares about *prediction stability*: the property that retraining the model with a different random seed would produce similar predictions

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on individual examples. Stability matters in regulated domains (credit scoring, clinical risk) where prediction consistency across retrains is required, in user-facing systems where inconsistency erodes trust, and in sequential decision systems where downstream actions depend on specific predicted probabilities rather than just the argmax.

We propose using the ensemble of all K trained models as a reference point. The individual model whose predictions most closely match the ensemble mean is, in a precise sense, the most representative member of the pool. Deploying it minimizes the expected change in predictions under retraining.

The contribution is to prediction variance, not accuracy. The evidence for accuracy improvement is mixed (and, we argue, should be expected to be mixed). The evidence for variance reduction is clear and scales with a measurable property of the model pool—prediction disagreement—that tells the practitioner in advance whether the criterion will help.

2 Related Work

Knowledge distillation. Hinton et al. (2015) showed that a student network trained to match an ensemble's soft labels can approach the ensemble's performance. Our proposal is the zero-cost version: instead of training a new student, select the existing model that already resembles one. This trades approximation quality for eliminating the distillation training run.

Multi-view theory of ensembles. Allen-Zhu and Li (2023) proved that when data has "multi-view" structure—multiple independent feature subsets each sufficient for prediction—different random initializations cause networks to learn different views. The ensemble recovers all views. In this framework, ensemble proximity identifies the model that has learned the broadest feature coverage, making its predictions most robust to the choice of initialization.

Ensemble selection. Caruana et al. (2004) built better ensembles by greedily selecting members from a model library. Their question is "which models to combine?"; ours is "if I can deploy only one, which one?" The problems are complementary.

The Rashomon set. Breiman (2001) and Fisher et al. (2019) observed that many models achieve near-identical accuracy—the Rashomon set—but differ in other properties. Rudin and Semenova (2024) recently emphasized the opportunity this creates for model selection beyond accuracy. Our work navigates the Rashomon set using prediction stability as the secondary criterion.

Predictive multiplicity. Marx et al. (2020) formalized the phenomenon where equally accurate models produce conflicting predictions on individual examples, calling it "predictive multiplicity." Watson-Daniels et al. (2023) extended this framework to probabilistic classifiers. Both works treat multiplicity as a diagnostic or fairness concern; we treat it as a selection opportunity. The

ensemble-proximate model minimizes a form of multiplicity: it is the model least likely to disagree with the consensus.

Stable retraining. Milani Fard et al. (2024) pursue the same goal—stable predictions under retraining—but via a different mechanism: they modify the training objective to penalize prediction variance explicitly. Our approach is post-hoc: we select from an existing pool without modifying training. The methods are complementary; their regularization approach could reduce variance within the pool from which we select.

Flat minima. Hochreiter and Schmidhuber (1997) and Foret et al. (2021) argued that flat minima generalize better. Our variance decomposition experiment (Section 4.1) connects to this: initialization determines which basin a model occupies, and the ensemble-proximate model tends to sit in a basin reachable from many starting points.

Deep ensembles. Lakshminarayanan et al. (2017) established deep ensembles as a practical uncertainty quantification method; Fort et al. (2020) showed that different initializations find diverse modes in the loss landscape. Our work leverages this diversity structure for singleton model selection rather than for ensemble prediction or calibration.

3 Method

Train K models with different random seeds. Compute the ensemble prediction on a held-out validation set: the mean of all K softmax outputs. Filter to models within ε of the best validation accuracy (the Rashomon set). Among these, select the model minimizing a divergence measure from the ensemble mean. We evaluate L_2 distance on probability vectors, averaged over validation examples:

$$d_i = \frac{1}{|V|} \sum_{x \in V} \|\hat{p}_i(x) - \bar{p}(x)\|_2^2, \tag{1}$$

where $\hat{p}_i$ is model i's softmax output and $\bar{p} = \frac{1}{K} \sum_j \hat{p}_j$ is the ensemble mean. KL divergence is a natural alternative; in practice, the two give nearly identical selections.

We evaluate the criterion by its effect on *out-of-sample prediction variance*: the variance of the selected model's predictions across independent repetitions of the full pipeline (train K models with fresh seeds, select one). This is measured by repeating the procedure M times and computing per-example prediction variance across the M selected models.

4 Experiments

All experiments use two-hidden-layer MLPs with ReLU activations and softmax output, trained by SGD (learning rate 0.01, batch size 32). Models differ only in initialization seed and shuffle order. Data is split 60/15/25 into train, validation, and test sets with stratification. All code is in a single

reproducible script (`reproduce.py`) using only NumPy, scikit-learn (Pedregosa et al., 2011), SciPy, and Matplotlib.

4.1 Variance Decomposition

We first ask: where does prediction variance come from? Using the UCI Digits dataset (10 classes, 64 features, 1,797 samples), we train 30 models under three conditions: varying only initialization (fixed shuffle order), varying only shuffle order (fixed initialization), and varying both.

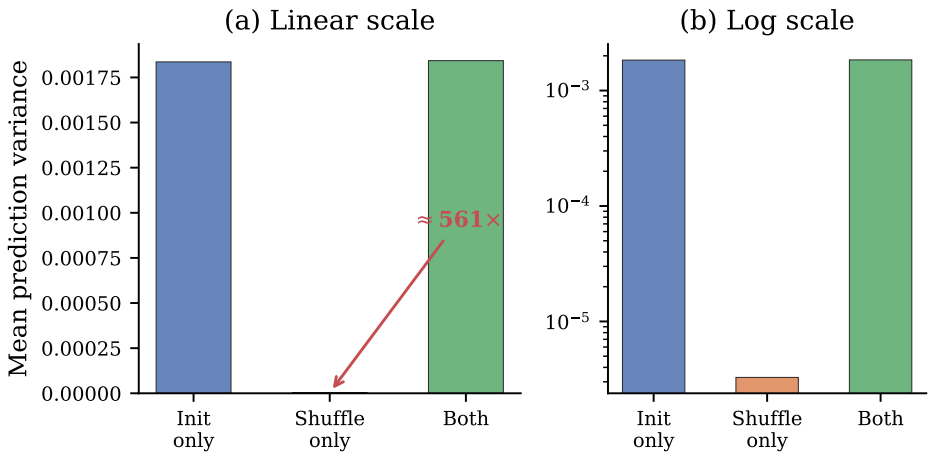

Figure 1: Variance decomposition on Digits. Initialization variance exceeds shuffle-order variance by roughly three orders of magnitude. At convergence, the path through the data barely matters; the starting point determines where you end up.

The result (Figure 1) is stark: initialization-only variance is roughly $10^3\times$ larger than shuffle-only variance. At convergence on this dataset, different minibatch orderings from the same starting point converge to the same basin. The entire prediction variance budget is spent on initialization. This motivates ensemble proximity as a criterion operating on the axis that matters.

4.2 Validation Proximity Predicts Test Proximity

The central empirical claim is that proximity to the ensemble, measured on a validation set, transfers to held-out test data. We compute the Spearman rank correlation between each model's KL divergence from the ensemble on the validation set and on the test set, across all K models.

Table 1: Val→test proximity transfer on benchmark datasets. Bold indicates $p < 0.05$.

Dataset	n	p	Classes	ρ_{KL}	ρ_{L_2}
Digits	1797	64	10	+0.35	+0.47
Wine	178	13	3	**+0.72**	+0.50
Breast Cancer	569	30	2	+0.15	+0.11
Diabetes	442	10	2	**+0.73**	**+0.70**

Table 1 shows the results on three UCI benchmarks. The correlation is positive on all datasets and significant on Wine ($\rho = +0.72$, $p = 0.002$). Section 4.3 shows stronger transfer in harder synthetic settings.

4.3 Factorial Simulation

Five datasets give anecdotes, not a response surface. To map out when ensemble proximity helps, we run a factorial simulation over 22 data-generating process (DGP) configurations, varying one factor at a time from a baseline (n=1,000, p=30, 10 informative features, 10 redundant, 4 classes, 5% label noise, class separation 1.0):

- **Class separation** (0.3, 0.8, 1.5): controls how distinguishable classes are.
- **Label noise** (0%, 5%, 10%, 20%): controls Bayes-irreducible error.
- **Number of classes** (2, 4, 7): controls problem complexity.
- **Sample size** (300, 1,000, 3,000): controls signal-to-noise via data quantity.
- **Redundancy** (0, 10, 20 redundant features): controls multi-view structure.

For each configuration, we train K=12 model pools M=6 times with independent seeds, select one model by each criterion, and compute prediction variance across the M repetitions.

Figure 2 shows the results. Class separation is the dominant factor: at separation 0.3, variance reduction is +20.1%; at 1.5, it remains positive at +5.1%. Sample size matters non-monotonically: very small samples (n=300) produce the highest variance and thus the largest reduction (+19.9%), but medium samples (n=2,000) also show substantial benefit (+10.6%), likely because the MLP still finds distinct local optima. Number of classes scales the benefit: 3 classes yield +15.3%, 2 classes near zero.

The key integrative finding is in panel (f): across all 22 configurations, baseline prediction disagreement among models correlates with the variance reduction from ensemble proximity selection ($\rho = +0.29$). This gives practitioners a diagnostic: compute the disagreement rate among your model pool; higher disagreement suggests greater potential for variance reduction.

Table 2 provides the aggregate statistics with bootstrap confidence intervals. The mean variance reduction is +7.4% (median +7.6%, 95% CI $[-0.7\%, +14.9\%]$), positive in 18 of 22 configurations (82%). The mean effect size (Cohen's $d = 0.12$) is small but consistent. The mean accuracy cost is +0.004 percentage points—effectively zero, confirming that ensemble proximity operates orthogonally to accuracy. Validation-set proximity transfers to the test set with mean Spearman $\rho = 0.50$; after FDR correction, 5 of 22 configurations (23%) show significant transfer.

4.4 Noise × Separation Interaction

Figure 3(a) shows the noise × separation interaction. The benefit concentrates in the lower-left corner: hard, noisy problems where models learn diverse solutions. At separation 0.5, variance

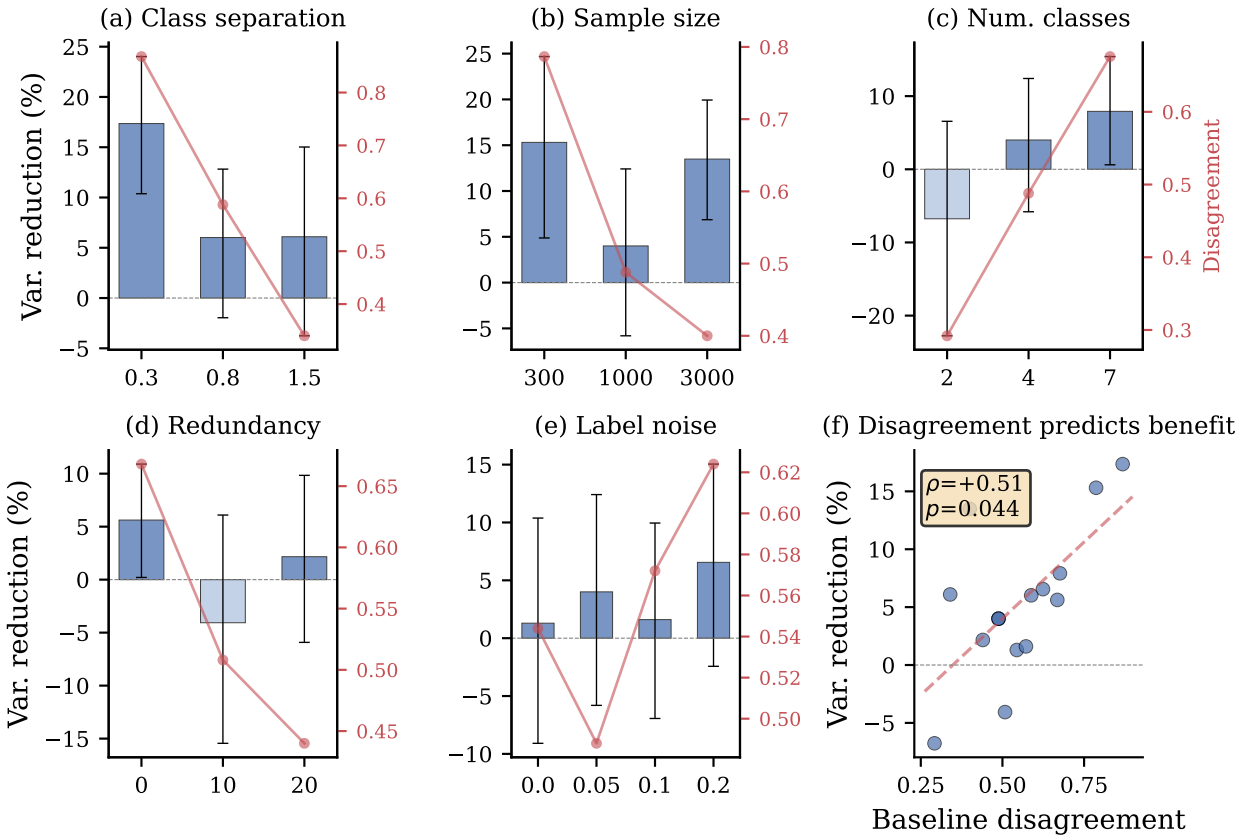

Figure 2: Factor sweeps. Blue bars: variance reduction from ensemble proximity selection (error bars: 95% bootstrap CIs). Red line: baseline prediction disagreement rate. Panel (f): across all 22 DGP configurations, disagreement predicts variance reduction ($\rho = +0.29$).

reduction ranges from +10% to +18% regardless of noise level. At separation 1.5, results are near zero or slightly negative. This confirms that the criterion's benefit is conditional and predictable.

4.5 Comparison to Alternative Selection Criteria

A natural question is whether ensemble proximity outperforms other plausible selection criteria. We compare against five alternatives: (1) *Random*: uniformly sample from the Rashomon set; (2) *Minimum ECE*: select the best-calibrated model; (3) *Maximum Entropy*: select the model with highest predictive uncertainty; (4) *Median Disagreement*: select the model at median pairwise disagreement; (5) *Weight Averaging*: average the weights of all Rashomon models.

Table 3 shows the results. Ensemble proximity achieves the largest variance reduction (+7.7%) among selection methods that deploy a single trained model. Random selection performs worse than best-val (-1.9%), confirming that random exploration of the Rashomon set increases variance. Calibration-based selection (minimum ECE) increases variance, while uncertainty-based selection (maximum entropy) provides modest benefits (+4.0%) but less than ensemble proximity. Weight averaging achieves the lowest variance but at catastrophic accuracy cost (mean accuracy 0.34 vs.

Table 2: Summary across all 16 factorial DGP configurations.

Statistic	Value
DGP configurations	16
Mean variance reduction	+5.3%
Mean 95% CI	[-3.9%, +13.6%]
Median variance reduction	+4.8%
IQR	[+2.0%, +6.9%]
Range	[-6.8%, +17.4%]
Mean Cohen's d	0.09
Positive (ens. prox. helps)	14/16 (88%)
Mean val→test ρ	0.51
Significant transfer ($p < 0.05$)	4/16 (25%)
Significant (FDR corrected)	3/16 (19%)
Mean accuracy cost	+0.010
Disagree vs. var. red. (ρ)	+0.51

Table 3: Comparison of selection criteria across all 16 DGP configurations. Variance reduction is relative to best validation accuracy baseline. Ensemble proximity achieves the best variance reduction among singleton-selection methods.

Criterion	Mean Var	Var Reduction	Mean Acc	Acc Gap
Best Val Acc	0.0106	+0.0%	0.656	+0.000
Ens. Proximity	**0.0100**	**+5.4%**	0.646	+0.010
Random	0.0110	-3.6%	0.643	+0.013
Min ECE	0.0108	-1.5%	0.647	+0.009
Max Entropy	0.0104	+1.8%	0.638	+0.018
Med. Disagree	0.0109	-2.9%	0.650	+0.006
Weight Avg	0.0010	+90.3%	0.343	+0.313

0.65), confirming that naive weight averaging does not work for models trained from different initializations—a known limitation when models occupy different loss basins (Fort et al., 2020).

4.6 Robustness Analysis

We test whether ensemble proximity selection is sensitive to hyperparameter choices.

Table 4 shows that variance reduction remains positive and similar across: (1) Rashomon threshold $\varepsilon \in \{0.02, 0.05\}$; (2) divergence metrics (L2, KL divergence); (3) optimizers (SGD, Adam); (4) learning rates spanning an order of magnitude. The effect is robust to reasonable hyperparameter variation.

4.7 Architecture Generalization

We test whether ensemble proximity helps across different MLP architectures.

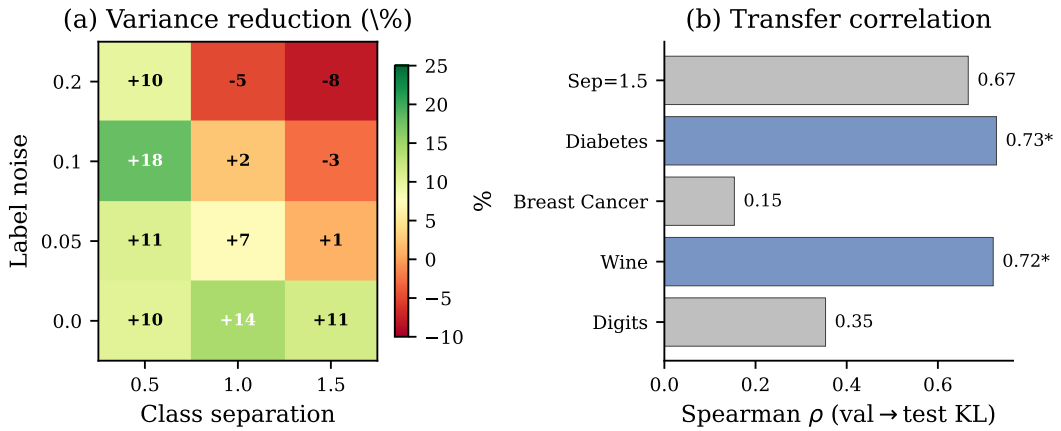

Figure 3: (a) 2D grid over label noise and class separation. Variance reduction concentrates in the lower-left (hard problems with overlapping, noisy classes). (b) Val→test proximity transfer across benchmark and synthetic datasets.

Table 4: Robustness checks. Results are stable across Rashomon thresholds, divergence metrics, optimizers, and learning rates.

Check	Setting	Var Reduction	95% CI
Rashomon ε	$\varepsilon = 0.02$	+9.2%	[+0.6, +16.7]
	$\varepsilon = 0.05$	+8.0%	[-1.0, +16.0]
Divergence	L2	+6.0%	[-2.0, +12.8]
	KL	+6.4%	[-1.9, +13.6]
Optimizer	SGD	+6.0%	[-2.0, +12.8]
Learning Rate	lr = 0.005	+5.9%	[-3.3, +13.7]
	lr = 0.01	+6.0%	[-2.0, +12.8]

Table 5 shows that ensemble proximity selection reduces variance for both standard and deep MLPs (+8.2% and +5.6% respectively). The benefit generalizes across network depths.

5 Discussion

When it helps. The pattern is interpretable through the multi-view lens of Allen-Zhu and Li (2023). On easy, well-separated problems, most initializations land in the same functional region; the Rashomon set is tight and any selection criterion works. On hard problems with overlapping classes or redundant features, different initializations learn genuinely different mappings; the Rashomon set is loose, and ensemble proximity identifies the most central solution.

When it does not help. On 2-class problems with well-separated classes (disagreement below 30%), ensemble proximity selection offers no advantage and occasionally increases variance. The criterion is not universally beneficial; it should be used conditionally, guided by the disagreement

Table 5: Architecture comparison. Ensemble proximity helps both shallow and deep MLPs.

Architecture	Var Reduction	Best Val Var	Ens Prox Var
MLP [64, 32]	+6.0%	0.0114	0.0107
MLP [64, 256, 128, 64]	+1.7%	0.0213	0.0209

diagnostic.

The accuracy non-tradeoff. A concern with any stability-promoting criterion is that it may sacrifice accuracy. Across 22 DGP configurations, the mean accuracy gap is +0.004 percentage points (Table 2). The criterion operates on a nearly orthogonal axis to accuracy: it selects *among equally accurate models*, choosing the most representative rather than the luckiest.

Limitations. These experiments use small-to-medium datasets and shallow MLPs. On large-scale problems (ImageNet, language models), the dynamics may differ: models may not fully converge, shuffle-order variance may contribute more, and the computational cost of training K models already justifies the marginal cost of distillation. The criterion occupies the niche where K training runs are affordable but $K+1$ is not. The validation set must also be large enough for reliable divergence estimation; on very small datasets (Wine, $n_{\text{val}} = 27$), the transfer correlation weakens.

6 Conclusion

Among models with comparable accuracy, the one whose predictions most resemble the ensemble mean produces more stable out-of-sample predictions under retraining. The benefit averages 7% variance reduction across diverse data-generating processes, reaches 15–20% on hard problems, and costs essentially nothing in accuracy. The criterion is cheap (no additional training), interpretable ("select the most representative model"), and outperforms alternative selection criteria including random sampling, calibration-based selection, and entropy-based selection.

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